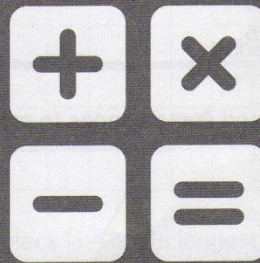




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**PAPER
A**



2011
ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

**Educational
Assessment
Australia**
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your
NAME, SCHOOL and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from
the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.
You are **NOT** allowed to use a calculator.

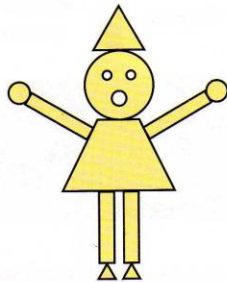
1.

Apples	
Oranges	
Bananas	
Pears	

How many oranges are shown in the table?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

2. Jess used some shapes to make this picture.



How many rectangles did she use?

- (A) 6
- (B) 4
- (C) 3
- (D) 1

3. Tony took three photos of the same branch.

Which of these is **NOT** one of Tony's photos?



(A)



(B)

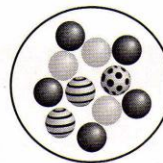


(C)



(D)

4. Natalia has the marbles shown below.



KEY

-  black
-  white
-  striped
-  spotted

Natalia mixes the marbles and picks up one of them without looking.

Which type of marble is she **least** likely to pick up?

- (A) black
- (B) white
- (C) striped
- (D) spotted

5. What time is shown on this clock?

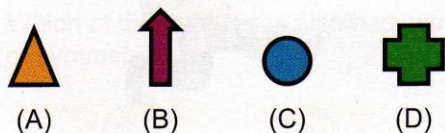


- (A) one past six
(B) six past one
(C) half past one
(D) six to two

6. Boris made a pattern of shapes. The picture shows the start of his pattern.



Which shape comes next in Boris's pattern?



7. The symbol  stands for the number of pets Anish has.

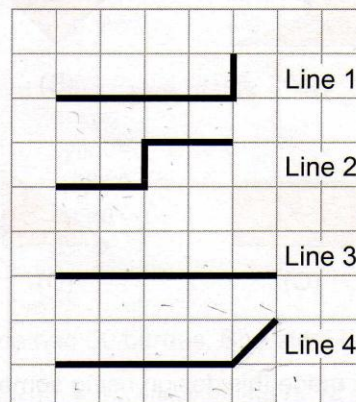
Tony has three more pets than Anish.

How many pets does Tony have?

- (A)  + 3
(B)  - 3
(C) 3 - 
(D) 3 × 

8. There are four black lines drawn on the grid.

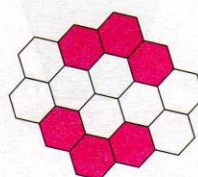
They are labelled Line 1, Line 2, Line 3 and Line 4.



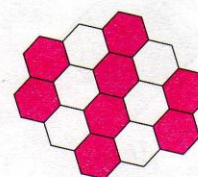
Which black line is the longest?

- (A) Line 1
(B) Line 2
(C) Line 3
(D) Line 4

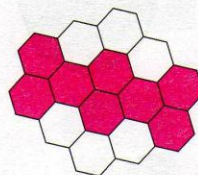
9. Which design has exactly half of its area shaded?



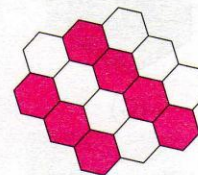
(A)



(B)

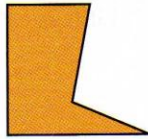


(C)



(D)

10. Which of these shapes has exactly two right angles?



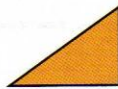
(A)



(B)

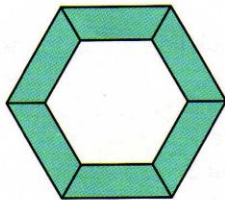


(C)

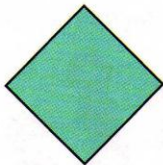


(D)

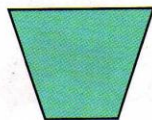
11. Jess made this design using some blocks.



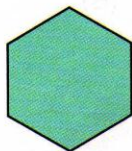
Which shape can fit exactly in the middle of her design without leaving any gaps?



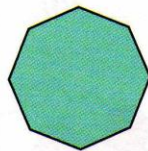
(A)



(B)

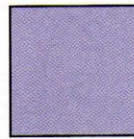


(C)



(D)

12. Which shape has the greatest area?



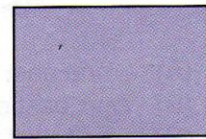
(A)



(B)



(C)



(D)

13. Tony left home at 8:30 am to go on a day trip. He arrived back home at 3:40 pm.



How long was Tony away from home?

- (A) 4 hours 50 minutes
(B) 5 hours 10 minutes
(C) 7 hours 10 minutes
(D) 11 hours 70 minutes
14. What is the number halfway between 20 and 46?
- (A) 26
(B) 30
(C) 33
(D) 36

15. Male bees live for about 45 days. Female bees live for about two months.



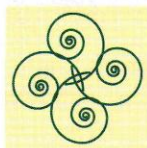
About how much longer than male bees do female bees live?

- (A) 15 days
- (B) 15 months
- (C) 43 days
- (D) 43 months

16. Celia drew this pattern. It has two lines of symmetry.



Which of these patterns also has two lines of symmetry?



(A)



(B)

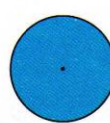


(C)



(D)

17. Here are two views of the same solid.



top view



side view

What is the name of this solid?

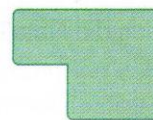
- (A) cylinder
- (B) sphere
- (C) prism
- (D) cone

18. Alice had 50 berries. She ate 17 berries.

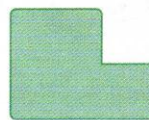
How many berries did she have left?

- (A) 33
- (B) 37
- (C) 43
- (D) 47

19. Anish drew this shape.



Which option shows the shape after a quarter turn clockwise?



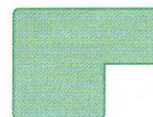
(A)



(B)

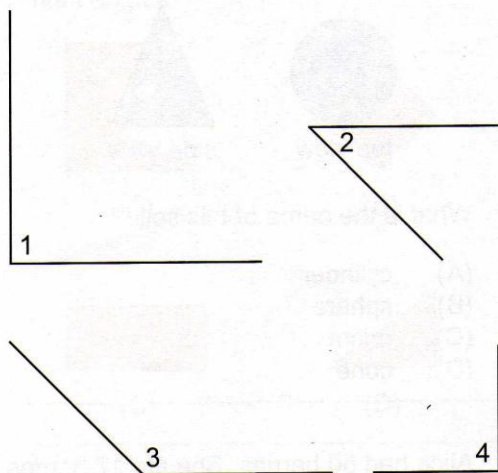


(C)



(D)

20. Which two angles are the same size?



- (A) 1 and 3
(B) 1 and 4
(C) 2 and 3
(D) 2 and 4

21. Natalia and Yara together drank a 500 mL bottle of juice. Yara drank four times as much juice as Natalia.

How much juice did Natalia drink, in mL?

- (A) 100
(B) 125
(C) 250
(D) 400

22. What is the missing number in this number pattern?

5, 11, 18, **?**, 35, 45, 56

- (A) 24
(B) 25
(C) 26
(D) 27

23. Jess and her friends were playing 'Pick Up Sticks'.

This table shows the points scored for picking up different coloured sticks.

Colour of stick	Point value
Yellow	1
Green	2
Blue	5
Red	10
Black	25

Jess picked up five yellow, five green, five blue and five red sticks.

What was the total point value of the sticks that Jess picked up?

- (A) 18
(B) 20
(C) 43
(D) 90

24. There are 50 students in Year 3.

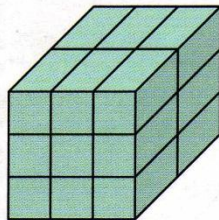
The table shows the number of Year 3 boys and girls in each sport group. Each student is in only one group.

Sport group	Girls	Boys
Red	3	9
Blue	8	7
Green	11	?

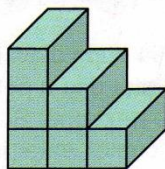
How many boys are in the Green group?

- (A) 38
(B) 34
(C) 16
(D) 12

25. Rachael made this shape from some small blocks.

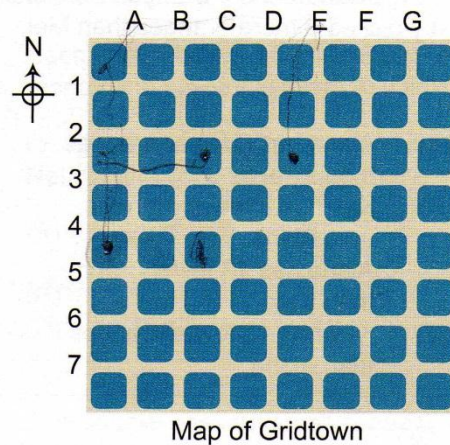


How many small blocks did she take away to make this shape?



- (A) 12
(B) 9
(C) 6
(D) 3

26. The picture shows a map of Gridtown.



Jack is at A5 and Debbie is at E3. Phil is two streets north of Jack and two streets west of Debbie.

Where is Phil?

- (A) C3
(B) G7
(C) A3
(D) E1

27. Tony and Anish tried out their toy cars to see which would go the longest distance.



Tony's car travelled five metres. Anish's car travelled 189 centimetres.

How much further than Anish's car did Tony's car travel?

- (A) 489 cm
(B) 411 cm
(C) 311 cm
(D) 184 cm

28. Alana, Mei and Kay were picking oranges. Alana picked 5 more oranges than Mei. Kay picked 8 more oranges than Mei.

Which of these is true?

- (A) Alana has 3 fewer oranges than Kay.
- (B) Kay has 3 fewer oranges than Alana.
- (C) Alana has 13 fewer oranges than Kay.
- (D) Kay has 13 fewer oranges than Alana.

29. Jess started listening to a CD when the minute hand on the clock was pointing to the 9.

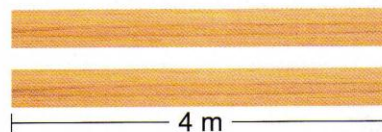
When she finished listening to the CD, the minute hand on the clock was pointing to the 5.



How long could Jess have listened to the CD?

- (A) 40 minutes
- (B) 20 minutes
- (C) 14 minutes
- (D) 4 minutes

30. Jess wants to build a house for her pet. She has two pieces of wood. Each piece is four metres long.



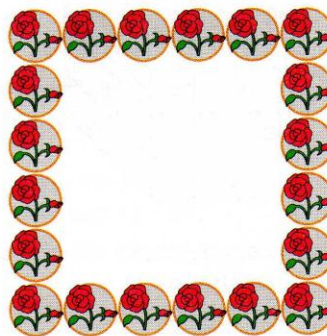
She needs to cut these pieces into shorter ones that are each 600 mm long.

What is the largest number of 600 mm pieces Jess can cut?

- (A) 14
- (B) 12
- (C) 7
- (D) 6

$$\begin{array}{r} 10 \\ 60 \overline{) 600} \end{array}$$

31. Natalia has 20 pots of roses as shown below.



Natalia had 94 cups of water. She poured one cup of water into a pot and then moved to the next one. She kept going, pouring one cup of water into each pot until there were no cups of water left.

Some pots got one cup of water more than others.

Which fraction of the pots got one more cup of water?

- (A) $\frac{3}{10}$
- (B) $\frac{5}{10}$
- (C) $\frac{7}{10}$
- (D) $\frac{8}{10}$

32. Tony has to take some medicine. He has to take two tablets, three times a day.

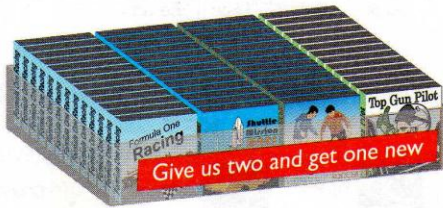
He has to take 32 tablets in total.

Today is the first day that Tony began taking the tablets.

When will Tony finish all the tablets?

- (A) on the fifth day
- (B) on the sixth day
- (C) on the eleventh day
- (D) on the sixteenth day

33. Anish traded all his old video games for new ones.



The game store gave Anish one new video game for every two old ones.

Then he was given four more new video games for his birthday.

Now Anish has ten new video games.

How many old video games did Anish have?

- (A) 3
- (B) 7
- (C) 12
- (D) 16

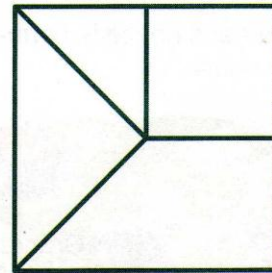
34. Each week Natalia receives \$30 pocket money and saves one-third of it.

Each week Tony receives \$16 pocket money and saves half of it.

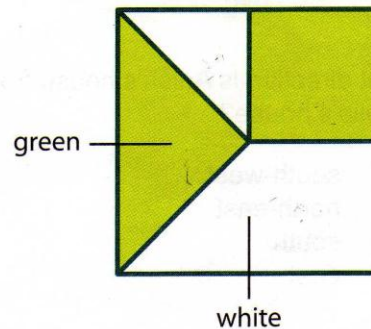
In eight weeks, how much money do Natalia and Tony save altogether?

- (A) \$368
- (B) \$144
- (C) \$88
- (D) \$18

35. This square has four small shapes.



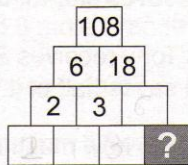
Jacob colours two of the small shapes green. Here is one way he could do this.



In how many different ways can Jacob colour two of the small shapes green?

- (A) 3
- (B) 6
- (C) 8
- (D) 9

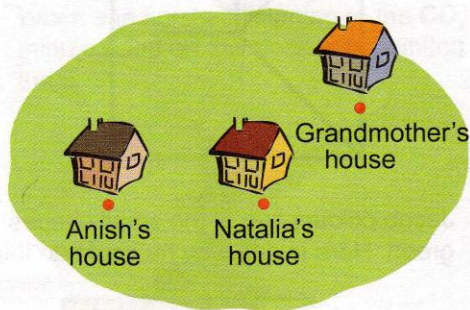
36. Here is a number pattern.



Which of these numbers should go into the shaded box to complete the pattern?

- (A) 2
(B) 3
(C) 10
(D) 12

37. Grandmother's house is north-west of Natalia's house.

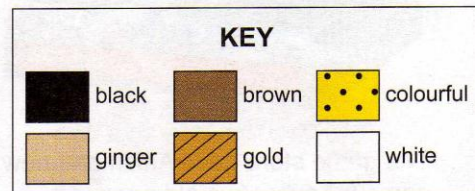
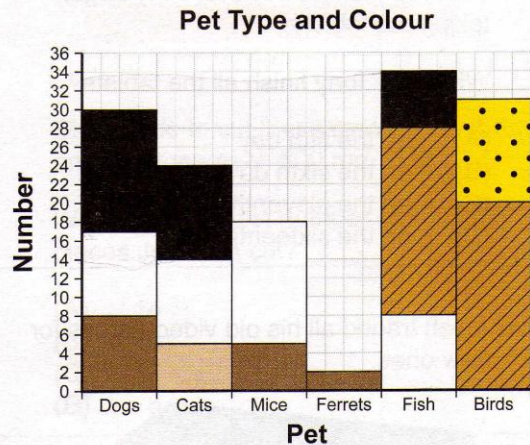


What direction is Anish's house from Natalia's house?

- (A) south-west
(B) north-east
(C) south
(D) east

38. Natalia recorded the type, colour and number of pets owned by students in her year at school.

She showed the results in a graph.



How many black pets were there in total?

- (A) 3
(B) 14
(C) 29
(D) 88

39. Which one of these fractions is the smallest?

(A) $\frac{4}{10}$

(B) $\frac{3}{4}$

(C) $\frac{1}{2}$

(D) $\frac{33}{100}$

40. Tony made this image using his computer.



He then flipped and turned the image.

Which one of these shows Tony's image after he flipped and turned it?



(A)



(B)



(C)



(D)



M

2011 Mathematics Answer Keys

ICAS
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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	B	A	B	B	A	A	C	A
2	B	D	C	D	B	D	B	D
3	D	B	D	C	B	D	A	C
4	D	A	B	C	C	D	D	A
5	C	A	A	B	C	A	A	D
6	C	B	C	D	B	A	C	A
7	A	C	D	A	A	B	A	B
8	D	D	C	D	B	D	C	B
9	D	B	D	A	D	A	D	C
10	A	B	D	A	D	B	A	C
11	C	D	A	D	D	D	D	A
12	D	C	A	D	C	A	B	A
13	C	B	B	C	B	A	A	C
14	C	C	B	D	A	A	B	B
15	A	B	B	A	A	D	D	C
16	B	D	A	A	D	B	A	B
17	D	A	A	B	A	C	B	D
18	A	D	D	A	D	C	C	D
19	B	A	D	A	C	C	B	A
20	B	C	A	C	C	B	B	C
21	A	D	D	B	D	A	A	C
22	C	D	C	B	B	C	C	C
23	D	B	A	C	C	B	B	B
24	D	D	C	C	B	A	C	D
25	A	A	C	B	D	B	B	B
26	A	B	D	B	B	B	B	C
27	C	A	A	C	A	B	C	B
28	A	D	B	C	D	C	A	B
29	A	D	B	B	A	B	A	D
30	B	C	D	D	D	C	C	B

(Please turn over)



M

2011 Mathematics Answer Keys

ICAS
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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
31	C	C	B	B	C	D	C	C
32	B	C	C	D	B	C	D	B
33	C	A	C	A	C	D	D	C
34	B	B	C	C	D	C	A	D
35	B	A	A	C	C	B	B	A
36	A	C	C	21 021	900	56 056	108	3 03 003 0 3
37	C	B	D	43 043	600	80 080	3 03 003 0 3	76 076
38	C	A	C	25 025	39 039	28 028	83 083	439
39	D	C	D	40 040	75 075	114	9 09 009 0 9	947
40	B	D	B	8 08 008 0 8	131	7 07 007 0 7	901	784

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